Retail Sales Performance Analysis

Comprehensive analysis of online retail transaction data using SQL, Power BI, and Python to uncover revenue patterns, customer behavior, and growth opportunities.

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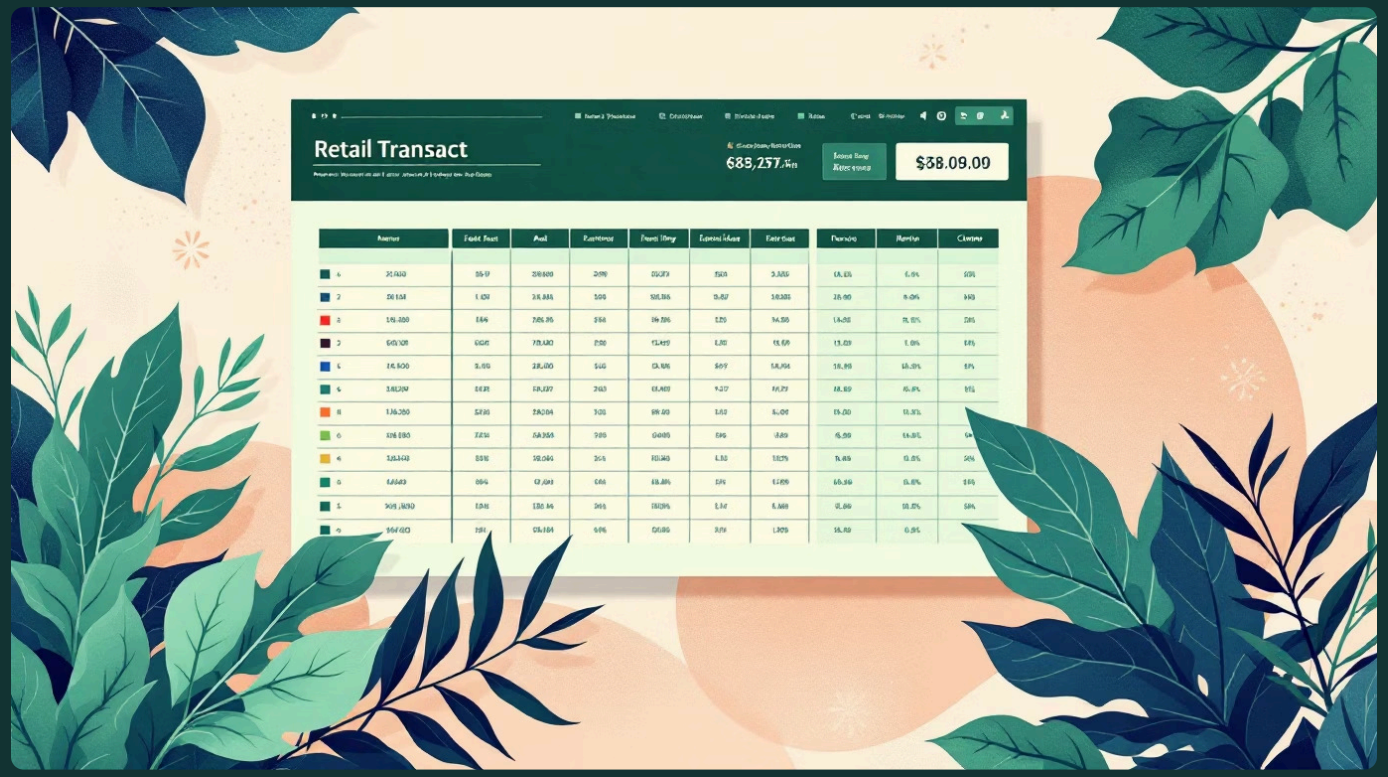
Dataset Overview

The dataset contains international online retail transactions with key fields including InvoiceNo, Description, Quantity, InvoiceDate, CustomerID, Country, and Revenue.

Each transaction represents a unique order with product details, customer information, and geographic location data.



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Key Business KPIs

$8.89M

Total Revenue

Generated from all transactions

18,536

Total Orders

Processed transactions

4,339

Total Customers

Unique customer base

$479.46

Average Order Value

High-value transactions

The business operates with high-value orders and a moderate customer base, indicating premium product positioning.

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Monthly Revenue Trends

1

Steady Growth

Consistent revenue through mid-year

2

Peak Season

September–November surge

3

Holiday Demand

Seasonal shopping behavior

Revenue increased significantly during September–November, driven by seasonal demand and holiday shopping. Retailers should prepare inventory and marketing campaigns during these peak months.

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Geographic Revenue Distribution

United Kingdom

$7.29M revenue from 3,921 customers

Secondary Markets

Netherlands, EIRE, Germany, France contribute smaller portions

Opportunity

Expand marketing in non-UK countries for revenue diversification

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Top Revenue-Generating Products

PAPER CRAFT

$168,470 revenue

REGENCY CAKESTAND

$142,265 revenue

WHITE HANGING HEART

$100,392 revenue

JUMBO BAG RED

$85,041 revenue

These high-demand products should maintain high inventory levels, feature in promotional campaigns, and bundle with complementary items.

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Customer Revenue Contribution

01

VIP Customers Identified

Small group contributes disproportionately large revenue

02

Top Performers

Customer 14646 generated $280,206; 18102 generated $259,657

03

Retention Strategy

Implement loyalty programs, personalized offers, priority service

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Weekly Sales Patterns

Peak Order Days

- Thursday: 4,033 orders
- Wednesday: 3,455 orders
- Tuesday: 3,185 orders

Orders peak toward end of week. Schedule promotions, discounts, and advertising campaigns during these peak days.

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Product Bundling Opportunities

Lunch Bag Combos

Red Retrospot frequently paired with Pink Polkadot and Black Skull designs

Teacup Collections

Green, Pink, and Roses Regency teacups purchased together

Light Holder Sets

Red and White Hanging Heart T-Light Holders bought together

Customers frequently purchase complementary products together, presenting bundling, cross-selling, and store layout optimization opportunities.

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Business Recommendations

1

Focus on High-Revenue Products

Promote best-selling items through targeted campaigns

2

Implement Product Bundling

Combine frequently purchased products to increase average order value

3

Retain High-Value Customers

Offer loyalty programs and exclusive discounts

4

Re-engage Inactive Customers

Run email marketing and promotional offers

5

Expand Geographic Markets

Increase marketing efforts in countries outside the UK

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